

Nikhil Singh

CONTACT INFORMATION	Room 515, Building G 26 Campus Kaiserslautern	<i>Contact:</i> niksingh@mpi-sws.org, +4916095870710 <i>GitHub:</i> https://github.com/ciniks117
CURRENT POSITION	Postdoctoral Researcher Max Planck Institute for Software Systems, Germany Supervisors : Prof. Sadegh Soudjani and Prof. Rupak Majumdar Member of AutoCyPher Project https://www.auto-cypher.eu/ September 2025 – Present	
EDUCATION	Indian Institute of Technology, Kanpur, PhD, Computer Science and Engineering, Jan 2020 - Sept 2024 (9.50/10) Thesis : Synthesis of Safe and Efficient Feedback Controllers for Complex Dynamical Systems. Advisor: Prof. Indranil Saha Indian Institute of Technology, Kanpur, MS, Computer Science and Engineering, Jan 2018 - Dec 2019 (9.0/10) Thesis : Specification guided Automated Debugging of CPS Models. Member of FMSAFE Project (MHRD IMPRINT) Advisor: Prof. Indranil Saha National Institute of Technology, Jamshedpur, B.Tech, Computer Science and Engineering, Aug 2010 - May 2014 (8.70/10) Project: Intelligent Web Search Engine.	
RESEARCH INTERESTS	Reinforcement Learning Robotics and Automation Formal Methods Control Theory	
EXPERIENCE	FARE Fellow, IIT Kanpur Safe Reinforcement Learning based Controller Synthesis. Visiting Researcher Max Planck Institute for Software Ssystems, Germany <i>Host: Prof. Rupak Majumdar</i> Safe Learning using Kernel based Embeddings. Software Engineer OptumSoft, Bangalore Software Development for Distributed Systems.	Jan'25 - Aug'25 Nov'24 - Jan'24 June'14 - Jun'16
PUBLICATIONS	* Rupak Majumdar, Nikhil Singh , Sadegh Soudjani. “ <i>Kernel-Based Safe Exploration in Deep Reinforcement Learning.</i> ” Learning for Dynamics & Control Conference (L4DC 2026), University of Southern California, (USC), Los Angeles, USA, June 17-19, 2026. * Rupak Majumdar, Mahmoud Salamati, Nikhil Singh , Sadegh Soudjani. “ <i>Learning Metastable Dynamics.</i> ” ACM International Conference on Hybrid Systems: Computation and Control, and ACM/IEEE International Conference on Cyber-Physical Systems, (HSCC/ICCPS 2026), Saint Malo, France, May 11-14 2026.	

Nikhil Singh, Indranil Saha. “*Reward Generation for Model-Free Reinforcement Learning from Formal Specifications.*” Journal of Artificial Intelligence Research (JAIR 2026).

Nikhil Singh, Indranil Saha. “*Frugal Actor-Critic: Sample Efficient Off-Policy Deep Reinforcement Learning Using Unique Experiences.*” International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2024), Auckland, New Zealand, May 06-10, 2024. <https://dl.acm.org/doi/10.5555/3635637.3663037>.

Code at <https://github.com/iitkcpslab/FAC>

Nikhil Singh, Indranil Saha. “*STL-Based Synthesis of Feedback Controllers Using Reinforcement Learning.*” AAAI Conference on Artificial Intelligence (AAAI 2023), Washington D.C., February 07-14, 2023. <https://ojs.aaai.org/index.php/AAAI/article/view/26764>.

Code at <https://github.com/iitkcpslab/rlstl>

Nikhil Singh, Indranil Saha. “*Specification Guided Automated Synthesis of Feedback Controllers.*” ACM SIGBED International Conference on Embedded Software (EMSOFT 2021), ACM/IEEE, virtual conference, October 08-15, 2021.

Published in ACM Transactions on Embedded Computing Systems (2021) <https://dl.acm.org/doi/10.1145/3477011>.

Code at <https://github.com/iitkcpslab/SCoSyn>

Nikhil Singh, Indranil Saha. “*Specification Guided Automated Debugging of CPS Models.*” International Conference on Embedded Software (EMSOFT2020), ACM/IEEE, virtual conference, September 20-25, 2020.

Published in IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (2020) <https://ieeexplore.ieee.org/document/9211574/>.

Code at <https://github.com/iitkcpslab/Blars>

* The author names appear in lexicographic order.

AWARDS & RECOGNITIONS

Fellowship for Academic and Research Excellence (Jan’25 - Aug’25).

Prime Minister’s Research Fellowship (Dec’20 - Dec-23).

ACM India IARCS travel grant (2024).

Invited to **Google Research Week 2023 & 2024**.

Nominated by the Department of CSE, IIT Kanpur for **Cadence Gold Medal** and **SIIC Student Innovation Award** for MS thesis.

First prize in Data structure & algorithm based event *CODATHON* in OJASS’13.

Third prize in Coding based event *CODEMANIA* in Ojass’ 13.

Second prize in the event *CODESTORM* conducted by the Computer Society of India at NIT-Jamshedpur held in March-2012.

First prize in the event *LET US C* conducted by the Computer Society of India at NIT-Jamshedpur held in April-2011.

TALKS

Kernel-Based Safe Exploration in Deep Reinforcement Learning
L4DC-26, Los Angeles, USA

Jun’26

Learning Metastable Dynamics
HSCC-26, Saint-Malo, France

May’26

Kernel-Based Safe Exploration in Deep Reinforcement Learning
Max Planck Institute for Software Systems, Germany

Apr’25

	Frugal Actor-Critic: Sample Efficient Off-Policy Deep Reinforcement Learning Using Unique Experiences <i>AAMAS-24, Auckland, New Zealand,</i>	May'24
	Learning-based Controller Synthesis <i>Ericsson AI Research</i>	Mar'23
	STL-Based Synthesis of Feedback Controllers Using Reinforcement Learning <i>AAAI-23, Washington D.C.</i>	Feb'23
	Specification Guided Automated Synthesis of Feedback Controllers <i>EMSOFT-21, Online</i>	Oct'21
	Specification-Guided Automated Debugging of CPS Models <i>EMSOFT-20, Online</i>	Sep'20
TEACHING	Instructor Introduction to PyTorch (PMRF Lecture series) <i>Institute of Smart Structures and Systems (ISSS)</i>	Sep'23 - Dec'23
	Instructor Controller Synthesis using Python (PMRF Lecture series) <i>Institute of Smart Structures and Systems (ISSS)</i>	Mar'23 - Jul'23
	Instructor Introduction to Controller Synthesis (PMRF Lecture series) <i>Institute of Smart Structures and Systems (ISSS)</i>	Sep'22 - Dec'22
	Tutor Introduction to Computing (ESC101) <i>Indian Institute of Technology, Kanpur</i>	May'22 - Jun'22
	Teaching Assistant Software Development (CS253) <i>Indian Institute of Technology, Kanpur</i>	Jan'22 - May'22
	Teaching Assistant Autonomous Cyber-Physical Systems (CS659A) <i>Indian Institute of Technology, Kanpur</i>	Jan'21 - May'21
	Teaching Assistant Embedded and Cyber-Physical Systems (CS637A) <i>Indian Institute of Technology, Kanpur</i>	Sept'20 - Dec'20
	Teaching Assistant Formal Methods for Robotics and Automation (CS638A) <i>Indian Institute of Technology, Kanpur</i>	Jan'20 - May'20
PROFESSIONAL SERVICES	PC Member ECAI-2025, AAAI-2025	
	Reviewer Conference : ICC-2024, IFAC-2026, HSCC-2026, IROS-2026	
	Journal: NAHS	